

Test yourself

Answers available from twalmsle@usc.edu

Q1: Write each of the following in matrix form and solve using Gauss-Jordan reduction:

- a. $2.x_1 - 3.x_2 = 2$ and $-4.x_1 + 2.x_2 = 3$
- b. $x_1 + x_2 = 1$, $x_2 + x_3 = 1$ and $2.x_1 + 3.x_2 - x_3 = 0$

Q2: Hoy et al, p299, Q1

For each of the matrix operations below, identify those that result in a scalar?

- a) AB where A is a 2×1 and B is a 1×2
- b) $A^T B$ where A is a 2×1 and B is a 2×1
- c) $A^T B A$ where A is a 2×1 and B is a 2×2
- d) $AA^T B$ where A is a 5×1 and B is a 5×5

Q3: Let

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & -2 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ -3 & 2 \end{bmatrix}, C = \begin{bmatrix} -1 & 1 & -3 \\ 0 & 2 & -2 \end{bmatrix} \text{ and } D = \begin{bmatrix} -1 & -2 & 0 \\ 1 & -2 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

Calculate each of the following if defined: $A+3B$, $A+C$, $C+2D$, AB , BA , CD , DC

Q4: Hoy et al, p299, Q2

Show that $AB \neq BA$ where:

$$A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$$

Q5: Hoy et al, p299, Q3

Let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \text{ and } x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Compute $(Ax)^T$, $x^T A^T$, xx^T , and $x^T x$. Is $A^T x^T$ defined?

Q6: Hoy et al, p345, Q1

Does the following matrix have an inverse?

$$\begin{bmatrix} 12 & 13 & 14 \\ 15 & 16 & 17 \\ 18 & 19 & 20 \end{bmatrix}$$

Q7: Determine the lengths of the column vectors:

$$u = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \text{ and } w = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

Are any of these vectors mutually perpendicular?

Q8: Hoy et al, p345, Q3

Use the co-factor expansion method to compute $|A|$, where:

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Q9: Hoy et al, p345, Q5

Use Cramer's rule to compute the solutions to

$$2x_1 + x_2 = 3$$

$$-3x_1 + x_3 = -8$$

$$x_2 + 2x_3 = -2$$

Q10: Hoy et al, p388, Q1

Establish whether the following pairs of vectors are linearly independent:

a) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

b) $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

c) $y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $z = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$

Q11: Hoy et al, p388, Q3

a) If A is a 9×7 matrix, what is the largest possible rank?

b) If A is a 7×9 matrix, what is the largest possible rank?

Q12: Hoy et al, p388, Q7

Find the eigenvalues of

a) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

b) $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$